



Yael Lallouz

TOTAL ROCKET FUEL

500



Certification Progress

ALPHA CERTIFIED PROGRESS: /24 COMPLETE
BETA CERTIFIED PROGRESS: Unlocks After Alpha Certification

Kindergarten

NY Coding	Standards Cod	Topic	Worksheets	In Progress	Certified	Month	Year
NY.K.CC.1	K.CC.1	Count to 100 by ones and by tens	☐	☐	☐		
NY.K.CC.2	K.CC.2	Count to 100 by ones beginning from any given number (instead of beginning at 1)	☐	☐	☐		
NY.K.CC.7	K.CC.7	Compare two written numerals between 1 and 10.	☐	☐	☐		
NY.K.OA.1	K.OA.1	Represent addition and subtraction with objects, fingers, mental images, drawings, sounds, acting out situations, verbal explanations, expressions, or equations.	☐	☐	☐		
NY.K.OA.2	K.OA.2	Solve addition and subtraction problems within 10 using objects or drawings to represent the problem.	☐	☐	☐		
NY.K.OA.3	K.OA.3	Decompose numbers up to 10 into pairs in more than one way and record each decomposition by a drawing or equation (e.g., $5 = 2 + 3$ and $5 = 4 + 1$).	☐	☐	☐		
NY.K.MD.1	K.MD.1	Describe measurable attributes of objects, such as length or weight. Describe several measurable attributes of a single object.	☐	☐	☐		
NY.K.MD.4	K.MD.4	Explore coins (pennies, nickels, dimes, and quarters) and begin identifying pennies and dimes	☐	☐	☐		
NY.K.G.1	K.G.1	Distinguish between defining attributes versus non-defining attributes for a wide variety of shapes. Build and/or draw shapes to possess defining attributes.	☐	☐	☐		
NY.K.G.3	K.G.3	Understand the difference between two-dimensional (lying in a plane, "flat") and three-dimensional ("solid") shapes.	☐	☐	☐		
NY.K.G.5	K.G.5	Model objects in their environment by building and/or drawing shapes	☐	☐	☐		
NY.K.G.6	K.G.6	Compose larger shapes from simple shapes	☐	☐	☐		

Kindergarten Notes

1st Grade

NY Coding	Standards Cod	Topic	Worksheets	In Progress	Certified	Month	Year
NY.1.NBT.1	1.NBT.1	Count to 120, starting at any number less than 120.	☐	☐	☐		
NY.1.OA.6	1.OA.6	Add and subtract within 20, demonstrating fluency for addition and subtraction within 10.	☐	☐	☐		
NY.1.OA.1	1.OA.1	Use addition and subtraction within 20 to solve word problems.	☐	☐	☐		
NY.1.OA.2	1.OA.2	Solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20.	☐	☐	☐		
NY.1.NBT.2	1.NBT.2	Understand that the two digits of a two-digit number represent amounts of tens and ones.	☐	☐	☐		
NY.1.NBT.3	1.NBT.3	Compare two two-digit numbers based on meanings of the tens and ones digits, recording the results of comparisons with the symbols $>$, $=$, and $<$.	☐	☐	☐		
NY.1.OA.3	1.OA.3	Apply properties of operations as strategies to add and subtract.	☐	☐	☐		
NY.1.OA.4	1.OA.4	Understand subtraction as an unknown-addend problem.	☐	☐	☐		
NY.1.OA.5	1.OA.5	Relate counting to addition and subtraction (e.g., by counting on 2 to add 2).	☐	☐	☐		
NY.1.OA.7	1.OA.7	Understand the meaning of the equal sign, and determine if equations involving addition and subtraction are true or false.	☐	☐	☐		
NY.1.OA.8	1.OA.8	Determine the unknown whole number in an addition or subtraction equation relating three whole numbers.	☐	☐	☐		
NY.1.NBT.4	1.NBT.4	Add within 100, including adding a two-digit number and a one-digit number, and adding a two-digit number and a multiple of 10.	☐	☐	☐		
NY.1.NBT.5	1.NBT.5	Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	☐	☐	☐		
NY.1.NBT.6	1.NBT.6	Subtract multiples of 10 in the range 10–90 from multiples of 10 in the range 10–90 (positive or zero differences).	☐	☐	☐		
NY.1.MD.1	1.MD.1	Order three objects by length; compare the lengths of two objects indirectly by using a third object.	☐	☐	☐		
NY.1.MD.2	1.MD.2	Express the length of an object as a whole number of length units, by laying multiple copies of a shorter object (the length unit) end to end; understand that the length measurement of an object is the number of same-size length units that span it with no gaps or overlaps.	☐	☐	☐		
NY.1.G.1	1.G.1	Distinguish between defining attributes versus non-defining attributes; build and draw shapes to possess defining attributes.	☐	☐	☐		
NY.1.G.2	1.G.2	Compose 2D shapes or 3D shapes to create other shapes	☐	☐	☐		
NY.1.G.3	1.G.3	Split circles and rectangles into two and four equal shares, describe the shares using the words halves, fourths, and quarters, and use the phrases half of, fourth of, and quarter of.	☐	☐	☐		
NY.1.MD.3	1.MD.3	Tell and write time and money.	☐	☐	☐		
NY.1.MD.3a	1.MD.3a	Tell and write time in hours and half-hours using analog and digital clocks. Develop an understanding of common terms, such as, but not limited to, o'clock and half past.	☐	☐	☐		
NY.1.MD.3b	1.MD.3b	Recognize and identify coins (penny, nickel, dime, and quarter) and their value and use the cent symbol (¢) appropriately.	☐	☐	☐		
NY.1.MD.3c	1.MD.3c	Count a mixed collection of dimes and pennies and determine the cent value (total not to exceed 100 cents).	☐	☐	☐		
NY.1.MD.4	1.MD.4	Organize, represent, and interpret data with up to three categories; ask and answer questions about the total number of data points, how many in each category, and how many more or less are in one category than in another.	☐	☐	☐		

1st Grade Notes

2nd Grade

2nd Grade Notes

NY Coding	Standards Cod	Topic	Worksheets	In Progress	Certified	Month	Year	
NY.2.OA.2a	2.OA.2a	https://rocketclub.com/takeabreak	☐	☐	☐			
NY.2.OA.2b	2.OA.2b	Know from memory all sums within 20 of two one-digit numbers	☐	☐	☐			
NY.2.MD.6	2.MD.6	Represent whole numbers as lengths from 0 on a number line with equally spaced points corresponding to the numbers 0, 1, 2, ..., and represent whole-number sums and differences within 100 on a number line	☐	☐	☐			
NY.2.MD.10	2.MD.10	Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and compare problems using information presented in a picture graph or a bar graph.	☐	☐	☐			
NY.2.OA.3a	2.OA.3a	Determine whether a group of objects (up to 20) has an odd or even number of members.	☐	☐	☐			
NY.2.NBT.4	2.NBT.4	Compare two three-digit numbers based on meanings of the hundreds, tens, and ones digits, using >, =, and < symbols to record the results of comparisons.	☐	☐	☐			
NY.2.OA.3b	2.OA.3b	Write an equation to express an even number as a sum of two equal addends	☐	☐	☐			
NY.2.OA.4	2.OA.4	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns. Write an equation to express the total as a sum of equal addends.	☐	☐	☐			
NY.2.OA.1a	2.OA.1a	Use addition and subtraction within 100 to solve one-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions.	☐	☐	☐			
NY.2.OA.1b	2.OA.1b	Use addition and subtraction within 100 to develop an understanding of solving two-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions.	☐	☐	☐			
NY.2.MD.1	2.MD.1	Measure the length of an object to the nearest whole by selecting and using appropriate tools such as rulers, yardsticks, meter sticks, and measuring tapes	☐	☐	☐			
NY.2.MD.2	2.MD.2	Measure the length of an object twice, using different "length units" for the two measurements; describe how the two measurements relate to the size of the unit chosen.	☐	☐	☐			
NY.2.MD.3	2.MD.3	Estimate lengths using units of inches, feet, centimeters, and meters.	☐	☐	☐			
NY.2.MD.4	2.MD.4	Measure to determine how much longer one object is than another, expressing the length difference in terms of a standard "length unit."	☐	☐	☐			
NY.2.MD.9	2.MD.9	Generate measurement data by measuring lengths of several objects to the nearest whole unit, or by making repeated measurements of the same object. Present the measurement data in a line plot, where the horizontal scale is marked off in whole-number units.	☐	☐	☐			
NY.2.MD.5	2.MD.5	Use addition and subtraction within 100 to solve word problems involving lengths that are given in the same units.	☐	☐	☐			
NY.2.NBT.1	2.NBT.1	Understand that the digits of a three-digit number represent amounts of hundreds, tens, and ones.	☐	☐	☐			
NY.2.NBT.1a	2.NBT.1a	Understand 100 can be thought of as a bundle of ten tens, called a "hundred."	☐	☐	☐			
NY.2.NBT.1b	2.NBT.1b	Understand the numbers 100, 200, 300, 400, 500, 600, 700, 800, 900 refer to one, two, three, four, five, six, seven, eight, or nine hundreds (and 0 tens and 0 ones).	☐	☐	☐			
NY.2.NBT.2	2.NBT.2	Count within 1000; skip-count by 5s, 10s, and 100s.	☐	☐	☐			
NY.2.NBT.3	2.NBT.3	Read and write numbers to 1000 using base-ten numerals, number names, and expanded form	☐	☐	☐			
NY.2.NBT.5	2.NBT.5	Fluently add and subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	☐	☐	☐			
NY.2.NBT.6	2.NBT.6	Add up to four two-digit numbers using strategies based on place value and properties of operations.	☐	☐	☐			
NY.2.NBT.7a	2.NBT.7a	Add and subtract within 1000, using • concrete models or drawings, and • strategies based on place value, properties of operations, and/or the relationship between addition and subtraction. Relate the strategy to a written representation	☐	☐	☐			
NY.2.NBT.7b	2.NBT.7b	Understand that in adding or subtracting up to three-digit numbers, one adds or subtracts hundreds and hundreds, tens and tens, ones and ones, and sometimes it is necessary to compose or decompose tens or hundreds.	☐	☐	☐			
NY.2.NBT.8	2.NBT.8	Mentally add 10 or 100 to a given number 100–900, and mentally subtract 10 or 100 from a given number 100–900.	☐	☐	☐			
NY.2.NBT.9	2.NBT.9	Explain why addition and subtraction strategies work, using place value and the properties of operations.	☐	☐	☐			
NY.2.G.1	2.G.1	Classify two-dimensional figures as polygons or non-polygons	☐	☐	☐			
NY.2.G.2	2.G.2	Partition a rectangle into rows and columns of same-size squares and count to find the total number of them.	☐	☐	☐			
NY.2.G.3	2.G.3	Partition circles and rectangles into two, three, or four equal shares. Describe the shares using the words halves, thirds, half of, a third of, etc. Describe the whole as two halves, three thirds, four fourths. Recognize that equal shares of identical wholes need not have the same shape.	☐	☐	☐			
NY.2.MD.7	2.MD.7	Tell and write time from analog and digital clocks in five minute increments, using a.m. and p.m. Develop an understanding of common terms, such as, but not limited to, quarter past, half past, and quarter to.	☐	☐	☐			
NY.2.MD.8a	2.MD.8a	Count a mixed collection of coins whose sum is less than or equal to one dollar	☐	☐	☐			
NY.2.MD.8b	2.MD.8b	Solve real world and mathematical problems within one dollar involving quarters, dimes, nickels, and pennies, using the € (cent) symbol appropriately.	☐	☐	☐			
3rd Grade								3rd Grade Notes
NY Coding	Standards Cod	Topic	Worksheets	In Progress	Certified	Month	Year	
NY.3.OA.1	3.OA.1	Interpret products of whole numbers.	☐	☐	☐			
NY.3.OA.2	3.OA.2	Interpret whole-number quotients of whole numbers.	☐	☐	☐			
NY.3.OA.7b	3.OA.7b	Know from memory all products of two one-digit numbers	☐	☐	☐			
NY.3.OA.9	3.OA.9	Identify and extend arithmetic patterns (including patterns in the addition table or multiplication table)	☐	☐	☐			
NY.3.MD.5	3.MD.5	Recognize area as an attribute of plane figures and understand concepts of area measurement.	☐	☐	☐			
NY.3.MD.5a	3.MD.5a	Recognize a square with side length 1 unit, called "a unit square," is said to have "one square unit" of area, and can be used to measure area	☐	☐	☐			
NY.3.MD.5b	3.MD.5b	Recognize a plane figure which can be covered without gaps or overlaps by n unit squares is said to have an area of n square units.	☐	☐	☐			
NY.3.MD.6	3.MD.6	Measure areas by counting unit squares.	☐	☐	☐			
NY.3.MD.7	3.MD.7	Relate area to the operations of multiplication and addition.	☐	☐	☐			
NY.3.MD.7a	3.MD.7a	Find the area of a rectangle with whole-number side lengths by tiling it, and show that the area is the same as would be found by multiplying the side lengths	☐	☐	☐			

NY.3.MD.7b	3.MD.7b	Multiply side lengths to find areas of rectangles with whole-number side lengths in the context of solving real world and mathematical problems, and represent whole-number products as rectangular areas in mathematical reasoning	☐	☐	☐		
NY.3.MD.7c	3.MD.7c	Use tiling to show in a concrete case that the area of a rectangle with whole-number side length a and side length b $\times$ c is the sum of $a \times b$ and $a \times c$. Use area models to represent the distributive property in mathematical reasoning.	☐	☐	☐		
NY.3.MD.7d	3.MD.7d	Recognize area as additive. Find areas of figures composed of non-overlapping rectangles, and apply this technique to solve real world problems.	☐	☐	☐		
NY.3.NBT.1	3.NBT.1	Use place value understanding to round whole numbers to the nearest 10 or 100	☐	☐	☐		
NY.3.NBT.2	3.NBT.2	Fluently add and subtract within 1,000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.	☐	☐	☐		
NY.3.NBT.4a	3.NBT.4a	Understand that the digits of a four-digit number represent amounts of thousands, hundreds, tens, and ones	☐	☐	☐		
NY.3.NBT.4b	3.NBT.4b	Read and write four-digit numbers using base-ten numerals, number names, and expanded form	☐	☐	☐		
NY.3.NBT.3	3.NBT.3	Multiply one-digit whole numbers by multiples of 10 in the range 10–90 using strategies based on place value and properties of operations.	☐	☐	☐		
NY.3.OA.3	3.OA.3	Use multiplication and division within 100 to solve word problems in situations involving equal groups, arrays, and measurement quantities.	☐	☐	☐		
NY.3.OA.4	3.OA.4	Determine the unknown whole number in a multiplication or division equation relating three whole numbers	☐	☐	☐		
NY.3.OA.5	3.OA.5	Apply properties of operations as strategies to multiply and divide.	☐	☐	☐		
NY.3.OA.6	3.OA.6	Understand division as an unknown-factor problem	☐	☐	☐		
NY.3.OA.7a	3.OA.7a	Fluently solve single-digit multiplication and related divisions, using strategies such as the relationship between multiplication and division or properties of operations.	☐	☐	☐		
NY.3.OA.8	3.OA.8	Solve two-step word problems posed with whole numbers and having whole-number answers using the four operations.	☐	☐	☐		
NY.3.OA.8a	3.OA.8a	Represent these problems using equations or expressions with a letter standing for the unknown quantity.	☐	☐	☐		
NY.3.OA.8b	3.OA.8b	Assess the reasonableness of answers using mental computation and estimation strategies including rounding.	☐	☐	☐		
NY.3.NF.1	3.NF.1	Understand a unit fraction, $1/b$, is the quantity formed by 1 part when a whole is partitioned into b	☐	☐	☐		
NY.3.NF.2	3.NF.2	Understand a fraction a as a number on the number line; represent fractions on a number line	☐	☐	☐		
NY.3.NF.2a	3.NF.2a	Represent a fraction $1/b$ on a number line by defining the intervals from 0 to 1 as the whole and partitioning it into b equal parts. Recognize that each part has size $1/b$ and that the endpoint of the part starting at 0 locates the number $1/b$ on the number line.	☐	☐	☐		
NY.3.NF.2b	3.NF.2b	Represent a fraction a/b on a number line by marking off a lengths $1/b$ from 0. Recognize that the resulting interval has size a/b and that its endpoint locates the number a/b on the number line.	☐	☐	☐		
NY.3.NF.3	3.NF.3	Explain equivalence of fractions and compare fractions by reasoning about their size.	☐	☐	☐		
NY.3.NF.3a	3.NF.3a	Understand two fractions as equivalent (equal) if they are the same size, or the same point on a number line.	☐	☐	☐		
NY.3.NF.3b	3.NF.3b	Recognize and generate equivalent fractions. Explain why the fractions are equivalent.	☐	☐	☐		
NY.3.NF.3c	3.NF.3c	Express whole numbers as fractions, and recognize fractions that are equivalent to whole numbers.	☐	☐	☐		
NY.3.NF.3d	3.NF.3d	Compare two fractions with the same numerator or the same denominator by reasoning about their size. Recognize that comparisons rely on the two fractions referring to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions	☐	☐	☐		
NY.3.MD.3	3.MD.3	Draw a scaled picture graph and a scaled bar graph to represent a data set with several categories. Solve one- and two-step "how many more" and "how many less" problems using information presented in a scaled picture graph or a scaled bar graph.	☐	☐	☐		
NY.3.MD.4	3.MD.4	Generate measurement data by measuring lengths using rulers marked with halves and fourths of an inch. Show the data by making a line plot where the horizontal scale is marked off in appropriate units—whole numbers, halves, or quarters.	☐	☐	☐		
NY.3.MD.2a	3.MD.2a	Measure and estimate liquid volumes and masses of objects using grams (g), kilograms (kg), and liters (l).	☐	☐	☐		
NY.3.MD.2b	3.MD.2b	Add, subtract, multiply, or divide to solve one-step word problems involving masses or liquid volumes that are given in the same units	☐	☐	☐		
NY.3.MD.1	3.MD.1	Tell and write time to the nearest minute and measure time intervals in minutes. Solve one-step word problems involving addition and subtraction of time intervals in minutes.	☐	☐	☐		
NY.3.G	3.G	Reason with shapes and their attributes.	☐	☐	☐		
NY.3.G.1	3.G.1	Recognize and classify polygons based on the number of sides and vertices (triangles, quadrilaterals, pentagons, and hexagons). Identify shapes that do not belong to one of the given subcategories.	☐	☐	☐		
NY.3.G.2	3.G.2	Partition shapes into parts with equal areas. Express the area of each part as a unit fraction of the whole.	☐	☐	☐		
NY.3.MD.8	3.MD.8	Geometric measurement: recognize perimeter as an attribute of plane figures and distinguish between linear and area measures	☐	☐	☐		
NY.3.MD.8a	3.MD.8a	Solve real world and mathematical problems involving perimeters of polygons, including finding the perimeter given the side lengths or finding one unknown side length given the perimeter and other side lengths.	☐	☐	☐		
NY.3.MD.8b	3.MD.8b	Identify rectangles with the same perimeter and different areas or with the same area and	☐	☐	☐		
4th Grade							
NY Coding	Standards Cod	Topic	Worksheets	In Progress	Certified	Month	Year
NY.4.OA.4	4.OA.4	Find all factor pairs for a whole number in the range 1–100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite	☐	☐	☐		
NY.4.NF.1	4.NF.1	Explain why a fraction a/b is equivalent to a fraction $x \times n/b \times n$ by using visual fraction models, with attention to how the number and size of the parts differ even though the two fractions themselves are the same size. Use this principle to recognize and generate equivalent fractions.	☐	☐	☐		

4th Grade Notes

NY.5.NF.4a	5.NF.4a	Interpret the product $a/b \times q$ as a parts of a partition of q into b equal parts; equivalently, as the result of a sequence of operations $a \times q \div b$.	☐	☐	☐		
NY.5.NF.4b	5.NF.4b	Find the area of a rectangle with fractional side lengths by tiling it with rectangles of the appropriate unit fraction side lengths, and show that the area is the same as would be found by multiplying the side lengths. Multiply fractional side lengths to find areas of rectangles, and represent fraction products as rectangular areas.	☐	☐	☐		
NY.5.NF.5	5.NF.5	Interpret multiplication as scaling (resizing).	☐	☐	☐		
NY.5.NF.5a	5.NF.5a	Compare the size of a product to the size of one factor on the basis of the size of the other factor, without performing the indicated multiplication	☐	☐	☐		
NY.5.NF.5b	5.NF.5b	Explain why multiplying a given number by a fraction greater than 1 results in a product greater than the given number (recognizing multiplication by whole numbers greater than 1 as a familiar case). Explain why multiplying a given number by a fraction less than 1 results in a product smaller than the given number. Relate the principle of fraction equivalence $a/b = a/b \times n/n$ to the effect of multiplying a/b by 1.	☐	☐	☐		
NY.5.NF.6	5.NF.6	Solve real world problems involving multiplication of fractions and mixed numbers.	☐	☐	☐		
NY.5.NF.7	5.NF.7	Apply and extend previous understandings of division to divide unit fractions by whole numbers and whole numbers by unit fractions.	☐	☐	☐		
NY.5.NF.7a	5.NF.7a	Interpret division of a unit fraction by a non-zero whole number, and compute such quotients.	☐	☐	☐		
NY.5.NF.7b	5.NF.7b	Interpret division of a whole number by a unit fraction, and compute such quotients.	☐	☐	☐		
NY.5.NF.7c	5.NF.7c	Solve real-world problems involving division of unit fractions by non-zero whole numbers and division of whole numbers by unit fractions.	☐	☐	☐		
NY.5.NBT.5	5.NBT.5	Fluently multiply multi-digit whole numbers using a standard algorithm.	☐	☐	☐		
NY.5.NBT.6	5.NBT.6	Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	☐	☐	☐		
NY.5.NBT.7	5.NBT.7	Using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between operations: • add and subtract decimals to hundredths; • multiply and divide decimals to hundredths. Relate the strategy to a written method and explain the reasoning used.	☐	☐	☐		
NY.5.OA.1	5.OA.1	Apply the order of operations to evaluate numerical expressions.	☐	☐	☐		
NY.5.OA.2	5.OA.2	Write simple expressions that record calculations with numbers, and interpret numerical expressions without evaluating them.	☐	☐	☐		
NY.5.OA.3	5.OA.3	Generate two numerical patterns using two given rules. Identify apparent relationships between corresponding terms. Form ordered pairs consisting of corresponding terms from the two patterns, and graph the ordered pairs on a coordinate plane.	☐	☐	☐		
NY.5.NBT.1	5.NBT.1	Recognize that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and 1/10 of what it represents in the place to its left.	☐	☐	☐		
NY.5.NBT.2	5.NBT.2	Use whole-number exponents to denote powers of 10. Explain patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of 10	☐	☐	☐		
NY.5.NBT.3	5.NBT.3	Read, write, and compare decimals to thousandths.	☐	☐	☐		
NY.5.NBT.3a	5.NBT.3a	Read and write decimals to thousandths using base-ten numerals, number names, and expanded form.	☐	☐	☐		
NY.5.NBT.3b	5.NBT.3b	Compare two decimals to thousandths based on meanings of the digits in each place, using >, =, and < symbols to record the results of comparisons.	☐	☐	☐		
NY.5.NBT.4	5.NBT.4	Use place value understanding to round decimals to any place.	☐	☐	☐		
NY.5.MD.1	5.MD.1	Convert among different-sized standard measurement units within a given measurement system when the conversion factor is given. Use these conversions in solving multi-step, real world problems.	☐	☐	☐		
NY.5.MD.2	5.MD.2	Make a line plot to display a data set of measurements in fractions of a unit (1/2, 1/4, 1/8). Use operations on fractions for this grade to solve problems involving information presented in line plots.	☐	☐	☐		
NY.5.MD.3	5.MD.3	Recognize volume as an attribute of solid figures and understand concepts of volume measurement.	☐	☐	☐		
NY.5.MD.3a	5.MD.3a	Recognize that a cube with side length 1 unit, called a "unit cube," is said to have "one cubic unit" of volume, and can be used to measure volume.	☐	☐	☐		
NY.5.MD.3b	5.MD.3b	Recognize that a solid figure which can be packed without gaps or overlaps using n unit cubes is said to have a volume of n cubic units.	☐	☐	☐		
NY.5.MD.4	5.MD.4	Measure volumes by counting unit cubes, using cubic cm, cubic in., cubic ft., and improvised units.	☐	☐	☐		
NY.5.MD.5	5.MD.5	Relate volume to the operations of multiplication and addition and solve real world and mathematical problems involving volume.	☐	☐	☐		
NY.5.MD.5a	5.MD.5a	Find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes, and show that the volume is the same as would be found by multiplying the edge lengths, equivalently by multiplying the height by the area of the base.	☐	☐	☐		
NY.5.MD.5b	5.MD.5b	Apply the formulas $V = l \times w \times h$ and $V = B \times h$ for rectangular prisms to find volumes of right rectangular prisms with whole-number edge lengths in the context of solving real world and mathematical problems.	☐	☐	☐		
NY.5.MD.5c	5.MD.5c	Recognize volume as additive. Find volumes of solid figures composed of two non-overlapping right rectangular prisms by adding the volumes of the non-overlapping parts, applying this technique to solve real world problems.	☐	☐	☐		
NY.5.G.1	5.G.1	Use a pair of perpendicular number lines, called axes, to define a coordinate system, with the intersection of the lines (the origin) arranged to coincide with the 0 on each line and a given point in the plane located by using an ordered pair of numbers, called its coordinates. Understand that the first number indicates how far to travel from the origin in the direction of one axis, and the second number indicates how far to travel in the direction of the second axis, with the convention that the names of the two axes and the coordinates correspond	☐	☐	☐		
NY.5.G.2	5.G.2	Represent real world and mathematical problems by graphing points in the first quadrant of the coordinate plane, and interpret coordinate values of points in the context of the situation.	☐	☐	☐		
NY.5.G.3	5.G.3	Understand that attributes belonging to a category of two-dimensional figures also belong to all subcategories of that category	☐	☐	☐		

NY.5.G.4	5.G.4	Classify two-dimensional figures in a hierarchy based on properties.	☐	☐	☐		
6th Grade							
NY Coding	Standards Cod	Topic	Worksheets	In Progress	Certified	Month	Year
NY.6.RP.1	6.RP.1	Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities.	☐	☐	☐		
NY.6.RP.2	6.RP.2	Understand the concept of a unit rate a/b associated with a ratio $a:b$ with $b \neq 0$ (b not equal to zero), and use rate language in the context of a ratio relationship.	☐	☐	☐		
NY.6.RP.3	6.RP.3	Use ratio and rate reasoning to solve real-world and mathematical problems.	☐	☐	☐		
NY.6.RP.3a	6.RP.3a	Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios.	☐	☐	☐		
NY.6.RP.3b	6.RP.3b	Solve unit rate problems.	☐	☐	☐		
NY.6.RP.3c	6.RP.3c	Find a percent of a quantity as a rate per 100. Solve problems that involve finding the whole given a part and the percent, and finding a part of a whole given the percent.	☐	☐	☐		
NY.6.RP.3d	6.RP.3d	Use ratio reasoning to convert measurement units; manipulate and transform units appropriately when multiplying or dividing quantities.	☐	☐	☐		
NY.6.NS.1	6.NS.1	Interpret and compute quotients of fractions, and solve word problems involving division of fractions by fractions.	☐	☐	☐		
NY.6.NS.2	6.NS.2	Fluently divide multi-digit numbers using a standard algorithm.	☐	☐	☐		
NY.6.NS.3	6.NS.3	Fluently add, subtract, multiply, and divide multi-digit decimals using a standard algorithm for each operation.	☐	☐	☐		
NY.6.NS.4	6.NS.4	Find the greatest common factor of two whole numbers less than or equal to 100. Use the distributive property to express a sum of two whole numbers 1–100 with a common factor as a multiple of a sum of two whole numbers with no common factor other than 1. Find the least common multiple of two whole numbers less than or equal to 12.	☐	☐	☐		
NY.6.NS.5	6.NS.5	Understand that positive and negative numbers are used together to describe quantities having opposite directions or values. Use positive and negative numbers to represent quantities in real-world contexts, explaining the meaning of 0 in each situation.	☐	☐	☐		
NY.6.NS.6	6.NS.6	Understand a rational number as a point on the number line. Use number lines and coordinate axes to represent points on a number line and in the coordinate plane with negative number coordinates.	☐	☐	☐		
NY.6.NS.6a	6.NS.6a	Recognize opposite signs of numbers as indicating locations on opposite sides of 0 on the number line. Recognize that the opposite of the opposite of a number is the number itself, and that 0 is its own opposite.	☐	☐	☐		
NY.6.NS.6b	6.NS.6b	Understand signs of numbers in ordered pairs as indicating locations in quadrants of the coordinate plane. Recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes.	☐	☐	☐		
NY.6.NS.6c	6.NS.6c	Find and position integers and other rational numbers on a horizontal or vertical number line. Find and position pairs of integers and other rational numbers on a coordinate plane.	☐	☐	☐		
NY.6.NS.7	6.NS.7	Understand ordering and absolute value of rational numbers.	☐	☐	☐		
NY.6.NS.7a	6.NS.7a	Interpret statements of inequality as statements about the relative position of two numbers on a number line.	☐	☐	☐		
NY.6.NS.7b	6.NS.7b	Write, interpret, and explain statements of order for rational numbers in real-world contexts.	☐	☐	☐		
NY.6.NS.7c	6.NS.7c	Understand the absolute value of a rational number as its distance from 0 on the number line. Interpret absolute value as magnitude for a positive or negative quantity in a real-world situation.	☐	☐	☐		
NY.6.NS.7d	6.NS.7d	Distinguish comparisons of absolute value from statements about order.	☐	☐	☐		
NY.6.NS.8	6.NS.8	Solve real-world and mathematical problems by graphing points on a coordinate plane. Include use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	☐	☐	☐		
NY.6.EE.1	6.EE.1	Write and evaluate numerical expressions involving whole-number exponents.	☐	☐	☐		
NY.6.EE.2	6.EE.2	Write, read, and evaluate expressions in which letters stand for numbers.	☐	☐	☐		
NY.6.EE.2a	6.EE.2a	Write expressions that record operations with numbers and with letters standing for numbers.	☐	☐	☐		
NY.6.EE.2b	6.EE.2b	Identify parts of an expression using mathematical terms (term, coefficient, sum, difference, product, factor, and quotient); view one or more parts of an expression as a single entity.	☐	☐	☐		
NY.6.EE.2c	6.EE.2c	Evaluate expressions given specific values for their variables. Include expressions that arise from formulas in real-world problems. Perform arithmetic operations, including those involving whole-number exponents, in the conventional order (Order of Operations).	☐	☐	☐		
NY.6.EE.3	6.EE.3	Apply the properties of operations to generate equivalent expressions	☐	☐	☐		
NY.6.EE.4	6.EE.4	Identify when two expressions are equivalent.	☐	☐	☐		
NY.6.EE.5	6.EE.5	Understand solving an equation or inequality as a process of answering a question: which values from a specified set, if any, make the equation or inequality true? Use substitution to determine whether a given number in a specified set makes an equation or inequality true.	☐	☐	☐		
NY.6.EE.6	6.EE.6	Use variables to represent numbers and write expressions when solving a real-world or mathematical problem. Understand that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set	☐	☐	☐		
NY.6.EE.7	6.EE.7	Solve real-world and mathematical problems by writing and solving equations of the form $x + p = q$; $x - p = q$; $px = q$; and $s/p = q$ for cases in which p , q , and x are all nonnegative rational numbers.	☐	☐	☐		
NY.6.EE.8	6.EE.8	Write an inequality of the form $x > c$, $x \geq c$, $x \leq c$, or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. Recognize that inequalities of these forms have infinitely many solutions; represent solutions of such inequalities on a number line.	☐	☐	☐		
NY.6.EE.9	6.EE.9	Use variables to represent two quantities in a real-world problem that change in relationship to one another. Given a verbal context and an equation, identify the dependent variable, in terms of the other quantity, thought of as the independent variable. Analyze the relationship between the dependent and independent variables using graphs and tables, and relate these to the equation.	☐	☐	☐		
NY.6.G.1	6.G.1	Find area of triangles, trapezoids, and other polygons by composing into rectangles or decomposing into triangles and quadrilaterals. Apply these techniques in the context of solving real-world and mathematical problems.	☐	☐	☐		
NY.6.G.2	6.G.2	Find volumes of right rectangular prisms with fractional edge lengths in the context of solving real-world and mathematical problems.	☐	☐	☐		

6th Grade Notes

NY.6.G.3	6.G.3	Draw polygons in the coordinate plane given coordinates for the vertices. Use coordinates to find the length of a side joining points with the same first coordinate or the same second coordinate. Apply these techniques in the context of solving real-world and mathematical problems.	☐	☐	☐		
NY.6.G.4	6.G.4	Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	☐	☐	☐		
NY.6.G.5	6.G.5	Use area and volume models to explain perfect squares and perfect cubes.	☐	☐	☐		
NY.6.SP.1a	6.SP.1a	Recognize that a statistical question is one that anticipates variability in the data related to the question and accounts for it in the answers	☐	☐	☐		
NY.6.SP.1b	6.SP.1b	Understand that statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population.	☐	☐	☐		
NY.6.SP.1c	6.SP.1c	Understand that the method and sample size used to collect data for a particular question is intended to reduce the difference between a population and a sample taken from the population so valid inferences can be drawn about the population. Generate multiple samples (or simulated samples) of the same size to recognize the variation in estimates or predictions	☐	☐	☐		
NY.6.SP.2	6.SP.2	Understand that a set of quantitative data collected to answer a statistical question has a distribution which can be described by its center, spread, and overall shape.	☐	☐	☐		
NY.6.SP.3	6.SP.3	Recognize that a measure of center for a quantitative data set summarizes all of its values with a single number while a measure of variation describes how its values vary with a single number.	☐	☐	☐		
NY.6.SP.4	6.SP.4	Display quantitative data in plots on a number line, including dot plots, and histograms.	☐	☐	☐		
NY.6.SP.5	6.SP.5	Summarize quantitative data sets in relation to their context.	☐	☐	☐		
NY.6.SP.5a	6.SP.5a	Report the number of observations	☐	☐	☐		
NY.6.SP.5b	6.SP.5b	Describe the nature of the attribute under investigation, including how it was measured and its units of measurement.	☐	☐	☐		
NY.6.SP.5c	6.SP.5c	Calculate range and measures of center, as well as describe any overall pattern and any striking deviations from the overall pattern with reference to the context in which the data were gathered.	☐	☐	☐		
NY.6.SP.5d	6.SP.5d	Relate the range and the choice of measures of center to the shape of the data distribution and the context in which the data were gathered.	☐	☐	☐		
NY.6.SP.6	6.SP.6	Understand that the probability of a chance event is a number between 0 and 1 inclusive, that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around $\frac{1}{2}$ indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event	☐	☐	☐		
NY.6.SP.7	6.SP.7	Approximate the probability of a simple event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability.	☐	☐	☐		
NY.6.SP.8	6.SP.8	Develop a probability model and use it to find probabilities of simple events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy.	☐	☐	☐		
NY.6.SP.8a	6.SP.8a	Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of simple events.	☐	☐	☐		
NY.6.SP.8b	6.SP.8b	Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process.	☐	☐	☐		

7th Grade							
NY Coding	Standards Cod	Topic	Worksheets	In Progress	Certified	Month	Year
NY.7.RP.1	7.RP.1	Compute unit rates associated with ratios of fractions.	☐	☐	☐		
NY.7.RP.2	7.RP.2	Recognize and represent proportional relationships between quantities.	☐	☐	☐		
NY.7.RP.2a	7.RP.2a	Decide whether two quantities are in a proportional relationship.	☐	☐	☐		
NY.7.RP.2b	7.RP.2b	Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.	☐	☐	☐		
NY.7.RP.2c	7.RP.2c	Represent a proportional relationship using an equation.	☐	☐	☐		
NY.7.RP.2d	7.RP.2d	Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points (0, 0) and (1, r) where r is the unit rate.	☐	☐	☐		
NY.7.RP.3	7.RP.3	Use proportional relationships to solve multistep ratio and percent problems.	☐	☐	☐		
NY.7.NS.1	7.NS.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers. Represent addition and subtraction on a horizontal or vertical number line.	☐	☐	☐		
NY.7.NS.1a	7.NS.1a	Describe situations in which opposite quantities combine to make 0.	☐	☐	☐		
NY.7.NS.1b	7.NS.1b	Understand addition of rational numbers; $p + q$ is the number located a distance $ q $ from p, in the positive or negative direction depending on whether q is positive or negative. Show that a number and its opposite have a sum of 0 (are additive inverses). Interpret sums of rational numbers by describing real-world contexts.	☐	☐	☐		
NY.7.NS.1c	7.NS.1c	Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in real-world contexts	☐	☐	☐		
NY.7.NS.1d	7.NS.1d	Apply properties of operations as strategies to add and subtract rational numbers.	☐	☐	☐		
NY.7.NS.2	7.NS.2	Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers.	☐	☐	☐		
NY.7.NS.2a	7.NS.2a	Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts.	☐	☐	☐		
NY.7.NS.2b	7.NS.2b	Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. If p and q are integers, then $-(p/q) = -p/q = p/-q$. Interpret quotients of rational numbers by describing real-world contexts.	☐	☐	☐		
NY.7.NS.2c	7.NS.2c	Apply properties of operations as strategies to multiply and divide rational numbers.	☐	☐	☐		
NY.7.NS.2d	7.NS.2d	Convert a fraction to a decimal using long division; know that the decimal form of a rational number terminates in 0s or eventually repeats.	☐	☐	☐		
NY.7.NS.3	7.NS.3	Solve real-world and mathematical problems involving the four operations with rational numbers.	☐	☐	☐		
NY.7.EE.1	7.EE.1	Work with linear expressions (add, subtract, factor, and expand) using rational numbers.	☐	☐	☐		
NY.7.EE.2	7.EE.2	Understand that rewriting an expression in different forms in a problem context can shed light on the problem and how the quantities in it are related.	☐	☐	☐		

7th Grade Notes

NY.7.EE.3	7.EE.3	Solve multi-step problems with positive and negative numbers (whole, fractions, decimals), using tools, converting between forms, and checking answers with estimation.	☐	☐	☐		
NY.7.EE.4	7.EE.4	Use variables to represent quantities in real-world problems and make simple equations or inequalities to solve them.	☐	☐	☐		
NY.7.EE.4a	7.EE.4a	Solve word problems with equations in the format $px + q = r$ or $p(x + q) = r$ where p, q, r are rational numbers, and compare algebraic and arithmetic methods of solving problems.	☐	☐	☐		
NY.7.EE.4b	7.EE.4b	Solve word problems with inequalities in the format $px + q > r$ or $px + q \leq r$, graph the solutions on a number line, and explain what they mean in the problem's context.	☐	☐	☐		
NY.7.G.1	7.G.1	Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	☐	☐	☐		
NY.7.G.2	7.G.2	Draw triangles when given measures of angles and/or sides, noticing when the conditions determine a unique triangle, more than one triangle, or no triangle.	☐	☐	☐		
NY.7.G.3	7.G.3	Describe the two-dimensional shapes that result from slicing three-dimensional solids parallel or perpendicular to the base.	☐	☐	☐		
NY.7.G.4	7.G.4	Apply the formulas for the area and circumference of a circle to solve problems.	☐	☐	☐		
NY.7.G.5	7.G.5	Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle in a figure.	☐	☐	☐		
NY.7.G.6	7.G.6	Solve real-world and mathematical problems involving area of two-dimensional objects composed of triangles and trapezoids. Solve surface area problems involving right prisms and right pyramids composed of triangles and trapezoids.	☐	☐	☐		
NY.7.SP.1	7.SP.1	Construct and interpret box-plots, find the interquartile range, and determine if a data point is an outlier.	☐	☐	☐		
NY.7.SP.3	7.SP.3	Informally assess the degree of visual overlap of two quantitative data distributions.	☐	☐	☐		
NY.7.SP.4	7.SP.4	Use measures of center and measures of variability for quantitative data from random samples or populations to draw informal comparative inferences about the populations.	☐	☐	☐		
NY.7.SP.8	7.SP.8	Find probabilities of compound events using organized lists, sample space tables, tree diagrams, and simulation.	☐	☐	☐		
NY.7.SP.8a	7.SP.8a	Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which the compound event occurs.	☐	☐	☐		
NY.7.SP.8b	7.SP.8b	Represent sample spaces for compound events using methods such as organized lists, sample space tables, and tree diagrams. For an event described in everyday language, identify the outcomes in the sample space which compose the event.	☐	☐	☐		
NY.7.SP.8c	7.SP.8c	Design and use a simulation to generate frequencies for compound events.	☐	☐	☐		